# The comparison of automatic artifact removal methods with robust classification strategies in terms of EEG classification accuracy

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***Abstract*—One of the key objectives of brain-computer interface (BCI) design is to construct accurate electroencephalogram (EEG) based classifier. But out of laboratory all EEG signals are contaminated with artifacts, which hamper algorithmic processing and EEG analysis, i.e. classifier ought to get a prediction for noisy data.**

**Real-time BCI system rely on relatively clean EEG signals. Therefore, the exclusion of artifacts is of special interest for BCI applications in everyday life. There are two main approaches to this objective: automatic EEG artifact rejection methods (subtract the noisy component) and robust classification methods (replace sensitive to outliers estimates with robust counterparts).**

**The goal of this work is to quantitatively compare popular automatic EEG artifact rejection approaches with robust classification methods in terms of motor imagery (MI) classification paradigm.**

## I. Introduction

Among the noninvasive brain research methods, the EEG method is the most popular. Due to the excellent temporal resolution EEG is widely used in BCI devices that can interpret brain activity and recognize the cognitive states [1]. Recently, portable BCI devices have appeared on the market. Nevertheless they are rather rare, because there is still a great challenge to construct accurate EEG based classifier. For example, classifier must be able to distinguish between right and left hand movements, or to evaluate the brain's response to a specific external stimulus using only recorded EEG data.

While the ideal way of working with EEG is to avoid the occurrence of artifacts when recording, out of laboratory all EEG signals are contaminated with artifacts, which hamper algorithmic processing and EEG analysis [2]. EEG artifacts also affect the classification accuracy. It means, that BCI classifier have to cope with EEG outliers too. In other words, BCI classifier have to get a prediction for noisy data and should get this prediction fast enough to be applied in real time applications.

There are two main approaches in the literature to deal with noisy EEG signal: automatic EEG artifact rejection methods and robust classification methods. First class of methods aims to subtract the noisy component from the EEG signal, while second class of methods aims to replace sensitive to outliers estimates with robust to outliers counterparts.

The goal of this work is to quantitatively compare popular automatic EEG artifact rejection approaches with robust classification methods in terms of EEG classification accuracy. In our research we want to understand, whether these methods provide improved classification accuracy and whether these methods can help in BCI software system design or they have no effect for real-time EEG classification. In order to compare proposed methods, MI classification paradigm is used. In the traditional MI paradigm a cue is given to the subject at the beginning of a trial. During each trial individual mentally simulates a given action, and action cue is also used as a label for this trial. Obviously, the performance of trained BCI system depends on the quality of training samples. In our research we use sensitive to outliers benchmark classification algorithm and noisy MI dataset.

This paper is organized as follows. Section II discusses in details benchmark solution for EEG classification problem in MI paradigm. Sections III and IV explain theory of automatic artifact rejection and robust classification methods, correspondingly. We conclude this work with dataset and performance criteria explanation as well as experiment results in section V. Summary and outlook are given in section VI.

## II. Benchmark solution for EEG classification problem in MI paradigm

To compare automatic artifact rejection methods with robust classification strategies, it is essential to select benchmark EEG classification algorithm. This algorithm should be unstable to outliers. In this way, we can measure the improvement of proposed methods over the classical benchmark algorithm and compare former methods between each other.

Benchmark algorithm consists of two main steps: feature extraction and classification. Respectively, multiple band-pass CSP algorithm for two class EEG problem and logistic regression classifier were used [3]. To extend this algorithm to more

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then two classes, one-versus-one feature engineering scheme is applied. We describe in details the process of benchmark algorithm learning for training dataset with $N$ formalized examples:

$$\{\boldsymbol{E}_i, y_i\}_{i=1}^{N},$$

where matrix $\boldsymbol{E}_i \in \mathbf{R}^{C\times T}$ represents EEG trial (channels, samples) and scalar $y_i \in \{1, \ldots, M\}$ belongs to one of $M$ classes.

### A. Multiple band-pass CSP

Multiple band-pass CSP feature extraction algorithm can operate only on two MI classes. It decomposes the EEG into multiple frequency pass bands using causal filter and applies CSP for each band-pass filtered range. For each band-pass filtered signal the CSP algorithm computes the transformation matrix $\boldsymbol{W}$ to yield features whose variances are optimal for discriminating two classes of EEG measurements. To do this, it solves the following optimization problem:

$$\begin{aligned} \max_{\boldsymbol{W}} \quad & \operatorname{tr} \boldsymbol{W}^T \boldsymbol{\Sigma}_1 \boldsymbol{W} \\ \text{s.t.} \quad & \boldsymbol{W}^T(\boldsymbol{\Sigma}_1 + \boldsymbol{\Sigma}_2)\boldsymbol{W} = \boldsymbol{I}, \end{aligned}$$

where intraclass mean covariance matrices can be calculated:

$$\boldsymbol{\Sigma}_1 = \underset{\boldsymbol{E}_i \in \{\text{class1}\}}{\text{mean}} \boldsymbol{E}_i \boldsymbol{E}_i^T / \operatorname{tr} \boldsymbol{E}_i \boldsymbol{E}_i^T$$

$$\boldsymbol{\Sigma}_2 = \underset{\boldsymbol{E}_i \in \{\text{class2}\}}{\text{mean}} \boldsymbol{E}_i \boldsymbol{E}_i^T / \operatorname{tr} \boldsymbol{E}_i \boldsymbol{E}_i^T$$

This problem can be solved by generalized eigenvalue value problem. After that, CSP algorithm selects best $m$ filters that can discriminate two classes of EEG measurements to use as:

$$\boldsymbol{W}_{CSP} = (\boldsymbol{w}_1, \ldots, \boldsymbol{w}_m) \in \mathbf{R}^{C\times m}$$

These filters maximize differences in variance between two MI classes. And for each new data trial $\boldsymbol{E}$ this feature extraction algorithm produces vector of features $\boldsymbol{x} = (x_1, \ldots, x_m)^T$ with components:

$$x_i = \log\left(\operatorname{var}[z_i] / \sum_{i=1}^{m} \operatorname{var}[z_i]\right),$$

where filtered signal matrix is given by $\boldsymbol{z} = (z_1, \ldots, z_m)^T$:

$$\boldsymbol{z} = \boldsymbol{W}_{CSP}^T \boldsymbol{E}$$

It should be noted, that multiple band-pass CSP feature extraction algorithm is sensitive to outliers because it involves the estimation of covariance matrices: classical sample covariance estimates are easily affected even if a single outlier exists.

### B. Multiple band-pass CSP with Riemannian mean

Multiple band-pass CSP, described above, use Euclidean mean to estimate intraclass covariance matrices. This approach is not completely correct, because covariance matrices of EEG signal are symmetric and positive-defined for proper data. They lie on a special manifold and, therefore, more accurate way to measure distance between two covariance matrices $\boldsymbol{C}_1, \boldsymbol{C}_2$ would be Riemannian distance [4]:

$$\delta_R(\boldsymbol{C}_1, \boldsymbol{C}_2) = \Big[\sum_{i=1}^{n} \log^2 \lambda_i\Big]^{1/2},$$

where $\lambda_1, \ldots, \lambda_n$ are the positive eigenvalues of $\boldsymbol{C}_1{}^{-1}\boldsymbol{C}_2$.

Riemannian mean replaces Euclidean for intraclass covariance matrix estimation. It can be written as

$$\mathfrak{G}(\boldsymbol{C}_1, \ldots, \boldsymbol{C}_N) = \underset{\boldsymbol{C} \in C(n)}{\arg\min} \sum_{i=1}^{N} \delta_R^2(\boldsymbol{C}, \boldsymbol{C}_i),$$

where $C(n)$ is the set of symmetric positive-defined matrices of shape $n * n$ with real values. The effective way to obtain Riemannian mean was described in [5].

### C. Classifier

On the second step of benchmark EEG classification algorithm logistic regression classifier without sparse penalization is used. Therefore, intermediate step with feature selection procedure is discarded and all features, calculated with multiple band-pass CSP, are applied as input for logistic classification rule. Logistic regression can handle multiclass problem. In our research one-versus-rest classification scheme is used.

## III. Automatic artifact rejection methods

First class of methods used to reject artifacts in EEG readings is based on assumption that electrode potentials are generated by mixing some components of pure brain activity with components of artifactual sources. And artifactual sources are concentrated in just a few independent components with structure that is strongly different from original brain activity [6]. To find this underlying components, blind source separation procedures are used.

Assume that signal $e_j$ can be described with linear mixture of $n$ independent components at each time $t$:

$$e_j = a_{j1}s_1 + \ldots + a_{jn}s_n,$$

where we assume that each mixture $e_j$ as well as each independent component $s_k$ is a random variable. In vector-matrix notation with random vector $\boldsymbol{e} = (e_1, \ldots, e_n)$ and random vector $\boldsymbol{s} = (s_1, \ldots, s_n)$ the above mixing model is written as:

$$\boldsymbol{e} = \boldsymbol{A}\boldsymbol{s},$$

where $\boldsymbol{A}$ the matrix with elements $a_{ij}$.

This statistical model is called independent component analysis, or ICA model. The independent components are latent variables, meaning that they cannot be directly observed. The starting point for ICA algorithm is the assumption that the components $s_i$ are statistically independent. Then, after estimating the matrix $\boldsymbol{A}$, we can compute its inverse $\boldsymbol{W}$, and obtain the independent component simply by:

$$\boldsymbol{s} = \boldsymbol{W}\boldsymbol{e}$$

After $\boldsymbol{W}$ calculation automatic artifact rejection methods use their heuristics to assess which of the found component is responsible for the original pure brain signal and which is artifactual signal. After this artifactual components are completely eliminated with their columns of the matrix of coefficients, cleared EEG signal is obtained.

In this research we used three state-of-the-art automatic artifact rejection algorithms, namely AAR [7,8], FASTER [9], and MARA [10,11]. Further details about heuristics used in these algorithms can be found in the original authors' papers. More information about existing ICA based artifact rejection heuristics can be found in original papers [12,13,14,15].

The main disadvantage of ICA based methodology is that the artifactual component may also contain neural activity aside from pure artifacts. In other words, the removal of the contaminated independent components, followed by a signal reconstruction may lead to distortions of the cerebral activity.

## IV. Robust classification methods

Robust classification methods are the second class of EEG artifact removal strategies. Robust methods can be used to detect outliers and reduce their influence. They utilize the idea that artifacts are spatially and temporally localized, and downweight effort of noisy samples and trials.

Benchmark feature extraction algorithm, multiple band-pass CSP, calculates covariance structure of each trial, which is sensitive to outliers. In order to provide robust estimates, proposed estimators do not downweight individual EEG samples, but rather reduce the influence of whole trials.

### A. Beta-divergence estimator

Classical covariance estimates are replaced by the robust covariance estimates. Estimator minimizes beta divergence between the empirical and a model Wishart distribution, thus allows to robustly average the estimated covariances of different trials and downweight the influence of outlier trials [16].

For each class with $N$ examples we have estimates of the trial covariance structures, known as the scatter matrices:

$$\boldsymbol{S}_i = \boldsymbol{E}_i \boldsymbol{E}_i^T \in \mathbf{R}^{C \times C}$$

To derive the average covariance matrix $\boldsymbol{\Sigma}$ from the trial covariances $\{\boldsymbol{S}_i\}_{i=1}^N$ algorithm minimizes beta divergence with Wishart distribution $q(\boldsymbol{S};\boldsymbol{\Sigma},\nu)$:

$$\frac{1}{2^{\frac{\nu C}{2}}|\boldsymbol{\Sigma}|^{\frac{\nu}{2}}\Gamma_C(\frac{\nu}{2})}|\boldsymbol{S}|^{\frac{\nu-C-1}{2}} \exp\left\{-\operatorname{tr}\frac{1}{2}\boldsymbol{\Sigma}^{-1}\boldsymbol{S}\right\},$$

where $\Gamma_C$ is the multivariate gamma function.

Whishart distribution is of great importance in the estimation of covariance matrices in multivariate statistics. A standard procedure to estimate this parameter is to maximize the log-likelihood. In robust manner it is equivalent to the minimization of so-called beta-divergence between the empirical and the model distribution. The beta divergence can be minimized by an iterative procedure [16]:

$$\boldsymbol{\Sigma}^{(k+1)} = \frac{\sum_{i=1}^N \psi_\beta(\boldsymbol{S}_i, \boldsymbol{\Sigma}^{(k)}, \nu)\boldsymbol{S}_i}{\nu \sum_{i=1}^N \psi_\beta(\boldsymbol{S}_i, \boldsymbol{\Sigma}^{(k)}, \nu) - \gamma |\boldsymbol{\Sigma}^{(k)}|^{\frac{(\nu-C-1)\beta}{2}}},$$

where

$$\psi_\beta(\boldsymbol{S}, \boldsymbol{\Sigma}, \nu) = |\boldsymbol{S}|^{\frac{(\nu-C-1)\beta}{2}} \exp\left\{-\operatorname{tr}\frac{\beta}{2}\boldsymbol{\Sigma}^{-1}\boldsymbol{S}\right\},$$

is a factor downweighting the influence of outlier trials and

$$\gamma = \frac{N\beta(C+1)}{2^{\frac{\nu C}{2}}\Gamma_C(\frac{\nu}{2})(\beta+1)}\left(\frac{2}{\beta+1}\right)^{\frac{\nu C(\beta+1)}{2}-\frac{C(C+1)\beta}{2}} \times$$
$$\times \Gamma_C\left(\frac{\nu(\beta+1)}{2} - \frac{(C+1)\beta}{2}\right)$$

Parameter $\beta$ determines the measure of robustness and should be selected carefully: very small $\beta$ values have no robustness effect whereas too large values have a too strong influence on the solution. To find the right trade-off is key for obtaining good performance.

## V. Methodology

This section describes in details EEG dataset used for classification and performance criteria as well as online classification methodology for real-time algorithm validation.

### A. MI data description

BCI Competition IV Dataset 2a consists of EEG data from 9 subjects [17]. The cue-based BCI paradigm consisted of four motor imagery tasks, namely the imagination of movement of the left hand, right hand, both feet, and tongue. For each subject there were two sessions, recorded on different days. Each session comprised 288 trials of data recorded with 22 EEG channels and 3 monopolar electrooculogram channels.

The typical features of this dataset are: balanced class distribution (the same number of trials for each of four classes), small number of electrodes, and the presence of artifacts, mostly expressed with ocular artifacts.

### B. Performance criteria

To analyze the performance of BCI systems, experiment confusion matrix is calculated for each algorithm. The confusion matrix for $M$-class classification problem shows the relationship between the true classes and the actual output of the classifier:

$$\left[n_{ij}\right] \in \mathbf{R}^{M \times M}$$

The elements $n_{ij}$ in the confusion matrix indicate how many samples of class $i$ have been predicted as class $j$. The total number of samples is $N = \sum_{i=1}^M \sum_{j=1}^M n_{ij}$.

In order to compare the performance of several algorithms on the same test dataset summary statistic is calculated. In this work algorithm performance was measured in terms of Cohen's kappa coefficient [18].

Cohen's kappa coefficient takes into account off-diagonal values, therefore controls not only overall agreement $p_0$:

$$p_0 = \frac{\sum_{i=1}^M n_{ii}}{N},$$

but chance agreement $p_e$ too:

$$p_e = \frac{\sum_{i=1}^M n_{:i} n_{i:}}{N^2},$$

where $n_{:i}$ and $n_{i:}$ – are the sum of the $i^{\text{th}}$ column and the $i^{\text{th}}$ row, respectively. Then, the estimate of the kappa coefficient $\kappa$ is:

$$\kappa = \frac{p_0 - p_e}{1 - p_e}$$

The kappa coefficient of a random classifier is 0 and kappa coefficient of 1 indicates perfect classification.

### C. Online paradigm and results

For each subject in BCI Competition IV Dataset 2a, using first session for train, classification algorithm should predict class for each sample from second session. This approach allows to simulate and evaluate the performance of the BCI in real-time and takes into account the speed of adaptation to the new EEG pattern.

Classification results are formulated in Table I.

TABLE I
COMPARISON OF CLASSIFICATION PERFORMANCES IN TERMS OF $\kappa$ FOR DIFFERENT ARTIFACT REJECTION ALGORITHMS (FROM LEFT TO RIGHT): BENCHMARK MULTIPLE BAND-PASS CSP, BENCHMARK MULTIPLE BAND-PASS CSP WITH RIEMANNIAN MEAN, AAR, FASTER, MARA, ROBUST CSP

| | BM1 | BM2 | AAR | FASTER | MARA | RCSP |
|---|---|---|---|---|---|---|
| S1 | 0.68 | 0.71 | 0.67 | 0.66 | 0.66 | 0.73 |
| S2 | 0.35 | 0.28 | 0.31 | 0.31 | 0.32 | 0.33 |
| S3 | 0.7 | 0.71 | 0.68 | 0.69 | 0.7 | 0.71 |
| S4 | 0.34 | 0.36 | 0.34 | 0.34 | 0.35 | 0.35 |
| S5 | 0.27 | 0.34 | 0.31 | 0.33 | 0.3 | 0.33 |
| S6 | 0.15 | 0.21 | 0.14 | 0.14 | 0.14 | 0.22 |
| S7 | 0.68 | 0.58 | 0.67 | 0.66 | 0.66 | 0.59 |
| S8 | 0.65 | 0.57 | 0.65 | 0.61 | 0.67 | 0.57 |
| S9 | 0.49 | 0.49 | 0.47 | 0.48 | 0.48 | 0.43 |
| AVG | 0.48 (0.2) | 0.47 (0.17) | 0.47 (0.19) | 0.47 (0.19) | 0.48 (0.2) | 0.47 (0.17) |

## VI. CONCLUSION

We have compared four different algorithms with two benchmark solutions in terms of EEG classification accuracy. From Table I there is no significant difference between proposed algorithms, but there is a large scatter of classification accuracy among subjects. Therefore, obtained results are not comparable. Although this dataset was positioned by organizers as highly contaminated, our experiments show that it is either not contaminated enough for mentioned algorithms or former algorithms aren't effective for this dataset.

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